

ELECTROMAGNETIC WAVES

①

DISPLACEMENT CURRENT-

Current that flows due to change in electric field is known as displacement current.

$$I = \frac{dq}{dt}$$

$$\phi_e = \frac{q}{\epsilon_0} \quad q = \phi_e \epsilon_0$$

$$I = \frac{d}{dt} \phi_e \epsilon_0$$

$$I_D = \epsilon_0 \frac{d\phi_e}{dt}$$

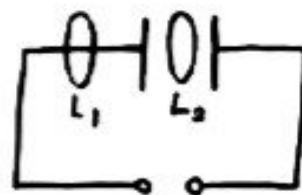
- The magnitude of I_D = magnitude of conduction current

MAXWELL'S MODIFICATION OF ACL

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_c + i_D)$$

$$\oint_{L_1} \vec{B} \cdot d\vec{l} = \mu_0 i_c \quad [\text{outside capacitor, } i_D = 0]$$

$$\oint_{L_2} \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi}{dt} \quad [\text{inside capacitor, } i_c = 0]$$



FOUR EQUATIONS OF ELECTROMAGNETISM

① Gauss theorem for electrostatics

→ To find flux and EF

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{en}}{\epsilon_0}$$

② Gauss theorem for magnetism

→ To find flux and MF

$$\oint \vec{B} \cdot d\vec{A} = 0$$

3. Maxwell's Ampere Circuital Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\phi_e}{dt} \right)$$

→ This proves changing electric flux creates MF.

4. Faraday's law of electromagnetic induction

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\phi_B}{dt}$$

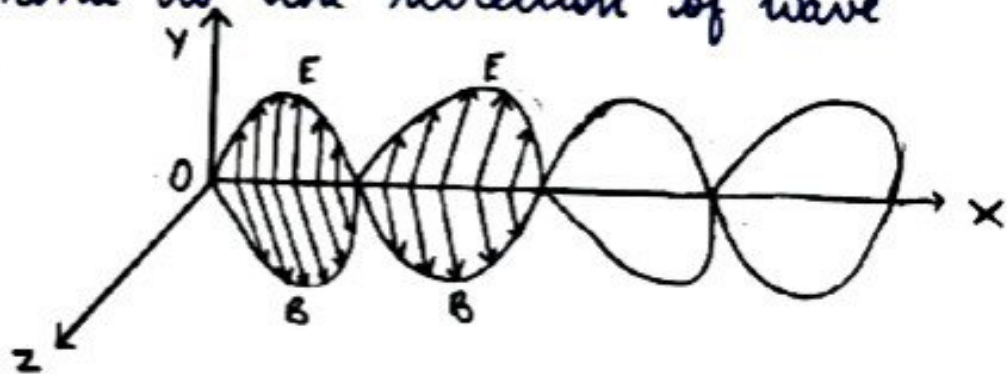
changing magnetic flux creates electric flux.

ELECTROMAGNETIC WAVES

A wave radiated by an accelerated or oscillatory charge in which varying magnetic field is the source of electric field and varying electric field is the source of magnetic field.

CHARACTERISTICS OF EM WAVES

1. The energy in EMW is divided on average equally between electric and magnetic fields.
2. The waves are transverse in nature.
3. EMW carry energy and exert force and pressure.
4. EMW are not deflected by electric & magnetic fields.
5. EMW are transverse in nature i.e. electric field and magnetic fields are perpendicular to each other and to the direction of wave propagation.



ELECTROMAGNETIC SPECTRUM

The systematic sequential distribution of EMW in ascending or descending order of frequency or wavelength is known as electromagnetic spectrum.

① Radio Wave -

- Wavelength range - $> 0.1 \text{ m}$
- Frequency range - $10^4 - 10^9 \text{ Hz}$
- Production - Rapid acceleration and deceleration of e^- .
- Detection - Receiver's aerials
- Uses - (i) In radio and TV communication
(ii) In astronomical field

② Microwaves

- Wavelength range - $0.1 \text{ m} - 1 \text{ mm}$
- Frequency range - $10^9 - 10^{11}$
- Production - Klystron valve or magnetron valve
- Detection - Point contact diodes
- Uses - (i) In RADAR communication
(ii) For cooking purpose

③ Infrared wave

- Wavelength range - $1 \text{ mm} - 700 \text{ nm}$
- Frequency range - $3 \times 10^{11} - 4 \times 10^{14}$
- Production - Vibration of atoms and molecules
- Detection - Thermopile, Bolometer
- Uses - (i) In treatment of muscular complaints
(ii) In knowing molecular structure

4. Visible rays

- Wavelength range - 700 nm to 400 nm
- Frequency range - 4×10^{14} - 8×10^{14} Hz
- Production - Electrons in atoms emit light when they move from one energy level to a lower energy level.
- Detection - The eye, photocells, photographic film
- Uses - (i) To see things
(ii) In optical instruments.

5. Ultraviolet Rays

- Wavelength range - 400 nm - 1 nm
- Frequency range - 8×10^{14} - 8×10^{16} Hz
- Production - Inner shell e^- in atoms moving from one energy level to a lower level.
- Detection - Photocells, photographic film
- Uses - (i) In burglar alarm
(ii) To kill germs in minerals

6. X-Rays

- Wavelength range - 1 nm - 10^{-3} nm
- Frequency range - 1×10^{16} - 3×10^{21}
- Production - X-ray tubes or inner shell e^-
- Detection - Photographic film, Geiger tubes
- Uses - (i) In medical diagnosis
(ii) In detecting faults, cracks

7. Gamma Rays

- Wavelength range - $< 10^{-3} \text{ nm}$
- Frequency range - $5 \times 10^{18} - 5 \times 10^{22} \text{ Hz}$
- Production - Radioactive decay of the nucleus
- Detection - Photographic film, ionisation chamber
- Uses - (i) For food preservation by killing pathogenic microorganisms.
(ii) In radiotherapy for treatment of tumour and cancer.